BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

CHEMISTRY (10th) — (Fresh/Re-appear) — (Version-B) — SECTION –A

Time Allowed: 20 Min | **Maximum Marks:** 12 | **Paper Code:** BN-X-XXIII-I

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The general formula of alkane is;

(A) C_nH_{2n}
(B) C_nH_{2n+2}
(C) C_nH_{2n-2}
(D) C_nH_{2n-2}

Q2. Deficiency of Vitamin-B causes;

(A) Lips inflammation
(B) Night blindness
(C) Rickets
(D) Hemorrhage

Q3. Triglycerides are the building blocks of;

(A) Proteins
(B) Carbohydrates
(C) Lipids
(D) Vitamins

Q4. All are AIR pollutants except;

(A) NO_2
(B) N_2
(C) SO_2
(D) CO

Q5. Water gas is a mixture of

(A) CO (g)+ H_2(g)
(B) C + H_2(g)
(C) CO + H_2O
(D) C + H_2O

Q6. The process in which the concentrated ore is heated either alone or with some other material in excess of air in a Furnace is called

(A) Smelting
(B) Roasting
(C) Bessemerization
(D) Electro-refining

Q7. The law of mass action was proposed in;

(A) 1862
(B) 1863
(C) 1864
(D) 1865

Q8. The bases which are soluble in water are called;

(A) Alkalis
(B) Salts
(C) Acids
(D) Amphoteric

Q9. Neutral solution has a pH value of;

(A) 4
(B) 7
(C) 10
(D) 14

Q10. Oxygen is bonded on both sides to carbon in;

(A) Alcohol
(B) Ketone
(C) Aldehyde
(D) Ether

Q11. Thiophene belongs to the organic class of;

(A) Ali-cyclic
(B) Aromatic
(C) Heterocyclic
(D) Homocyclic

Q12. Which one is less reactive;

(A) Ethene
(B) Ethyne
(C) Ethane
(D) Propene

CHEMISTRY (10th) — (Fresh/Re-appear) — SECTION –B

Time Allowed: 2:40Hrs | **Maximum Marks:** 32

> Note: Note: Attempt section B & C accordingly.

Q2. (Marks 32) Q.No 2. Attempt any EIGHT parts of the following. All parts carry equal marks.

i. Find out "Kc" and unit for the dynamic equilibrium. $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

ii. What is neutralization reaction? Support your answer with two examples.

iii. When a clear liquid is placed in a beaker, how can you identify it is an acid, base or neutral (water)?

iv. How Alkyl Radicals are formed? Discuss it with examples.

v. Compare molecular formula to structural formula.

vi. Why the colour of Bromine water discharges on addition to Ethene?

vii. Vitamins are vital for human life. Why?

viii. Write any four physical properties of Alkynes.

ix. How Ozone layer is depleted in the atmosphere by Nitric oxide (NO) and oxides of Sulphur?

x. Explain the importance of water and its quality.

xi. Assess the composition of urea and calculate the percentage of Nitrogen in it.

SECTION –C

Maximum Marks: 21

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) Explain the Lewis Acid Base concept with suitable examples.

b) Draw graph showing the rate of Forward and Reverse reaction and equilibrium state.

Q4.

a) How vital force theory was proved wrong? Give two important chemical reactions as well.

b) Write down examples of three unsaturated hydrocarbons with structural and condensed formulas.

Q5.

a) Write any four functions of Nucleic acids.

b) Mention the halogenation of adjacent Dihalides along with its two steps.

Q6.

a) Describe the chemical reactions that occur in the manufacture of urea in proper order.

b) Give the complete chemical reaction of H_2O with: i. $Cl_2(g)$ ii. C(s) iii. CaO(s)